Ex. No. 3	**Add Interactivity to App – Input and Forms**
Date of Exercise	
Github Code Link (Keep as public)	
Video Link (Google Drive Link)	

Aim

Develop a Flutter application that displays a predefined list of products and allows users to search or filter the products using input widgets, providing an interactive user experience.

Description

Interactivity in Flutter is commonly built on StatefulWidget combined with input widgets such as TextField. A TextEditingController gives programmatic access to whatever the user types, and the onChanged callback fires on every keystroke, letting the UI react immediately by calling setState to rebuild with new data.

This experiment implements a Product Search screen. A predefined, constant list of 14 products (name, category, price) is held in a model file. The SearchScreen is a StatefulWidget holding a TextEditingController and a _filtered list. Each keystroke in the TextField runs a case-insensitive substring match against both the product name and category and rebuilds the filtered list via setState. A clear ('x') button appears once text is entered, a live counter shows how many of the products currently match, and an explicit 'No products found' message is shown when the filter produces zero results, giving the user clear feedback at every step.

Program

lib/models/product.dart

```
class Product {
  final String name;
  final String category;
  final double price;

  const Product({required this.name, required this.category, required
this.price});
}

const List<Product> products = [
  Product(name: 'Wireless Headphones', category: 'Audio', price: 2499),
  Product(name: 'Smart Watch', category: 'Wearables', price: 3999),
  Product(name: 'Running Shoes', category: 'Footwear', price: 1899),
  Product(name: 'Coffee Maker', category: 'Home', price: 3299),
  Product(name: 'Backpack', category: 'Accessories', price: 1499),
  Product(name: 'Bluetooth Speaker', category: 'Audio', price: 1799),
```

```
  Product(name: 'Desk Lamp', category: 'Home', price: 899),
  Product(name: 'Sunglasses', category: 'Accessories', price: 1299),
  Product(name: 'Yoga Mat', category: 'Fitness', price: 999),
  Product(name: 'Water Bottle', category: 'Fitness', price: 499),
  Product(name: 'Mechanical Keyboard', category: 'Electronics', price: 2199),
  Product(name: 'Desk Chair', category: 'Home', price: 6499),
  Product(name: 'Wireless Mouse', category: 'Electronics', price: 799),
  Product(name: 'Hiking Boots', category: 'Footwear', price: 2899),
];
```

lib/widgets/product_tile.dart

```
import 'package:flutter/material.dart';

import '../models/product.dart';

class ProductTile extends StatelessWidget {
  final Product product;

  const ProductTile({super.key, required this.product});

  @override
  Widget build(BuildContext context) {
    return Card(
      margin: const EdgeInsets.symmetric(horizontal: 12, vertical: 5),
      child: ListTile(
        leading: CircleAvatar(child: Text(product.name[0])),
        title: Text(product.name),
        subtitle: Text(product.category),
        trailing: Text(
          '₹${product.price.toStringAsFixed(0)}',
          style: const TextStyle(fontWeight: FontWeight.bold),
        ),
      ),
    );
  }
}
```

lib/screens/search_screen.dart

```
import 'package:flutter/material.dart';

import '../models/product.dart';
import '../widgets/product_tile.dart';

class SearchScreen extends StatefulWidget {
  const SearchScreen({super.key});

  @override
  State<SearchScreen> createState() => _SearchScreenState();
}

class _SearchScreenState extends State<SearchScreen> {
  final TextEditingController _controller = TextEditingController();
  List<Product> _filtered = products;

  void _onSearchChanged(String query) {
    final q = query.trim().toLowerCase();
    setState(() {
      _filtered = q.isEmpty
          ? products
          : products
              .where((p) => p.name.toLowerCase().contains(q) ||
p.category.toLowerCase().contains(q))
              .toList();
    });
  }

  @override
  void dispose() {
```

```
    _controller.dispose();
    super.dispose();
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Product Search'), centerTitle: true),
      body: Column(
        children: [
          Padding(
            padding: const EdgeInsets.all(12),
            child: TextField(
              controller: _controller,
              onChanged: _onSearchChanged,
              decoration: InputDecoration(
                hintText: 'Search by name or category…',
                prefixIcon: const Icon(Icons.search),
                suffixIcon: _controller.text.isEmpty
                    ? null
                    : IconButton(
                        icon: const Icon(Icons.clear),
                        onPressed: () {
                          _controller.clear();
                          _onSearchChanged('');
                        },
                      ),
                border: OutlineInputBorder(borderRadius:
BorderRadius.circular(12)),
                filled: true,
              ),
            ),
          ),
          Padding(
            padding: const EdgeInsets.symmetric(horizontal: 16),
            child: Align(
              alignment: Alignment.centerLeft,
              child: Text(
                '${_filtered.length} of ${products.length} products',
                style: TextStyle(color: Colors.grey.shade600, fontSize: 13),
              ),
            ),
          ),
          const SizedBox(height: 4),
          Expanded(
            child: _filtered.isEmpty
                ? const Center(child: Text('No products found', style:
TextStyle(fontSize: 16)))
                : ListView.builder(
                    itemCount: _filtered.length,
                    itemBuilder: (context, index) => ProductTile(product:
_filtered[index]),
                  ),
          ),
        ],
      ),
    );
  }
}
```

lib/main.dart

```
import 'package:flutter/material.dart';

import 'screens/search_screen.dart';

void main() {
  runApp(const SearchApp());
}

class SearchApp extends StatelessWidget {
```

```
  const SearchApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Product Search',
      debugShowCheckedModeBanner: false,
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.teal),
        useMaterial3: true,
      ),
      home: const SearchScreen(),
    );
  }
}
```

Output Screenshots

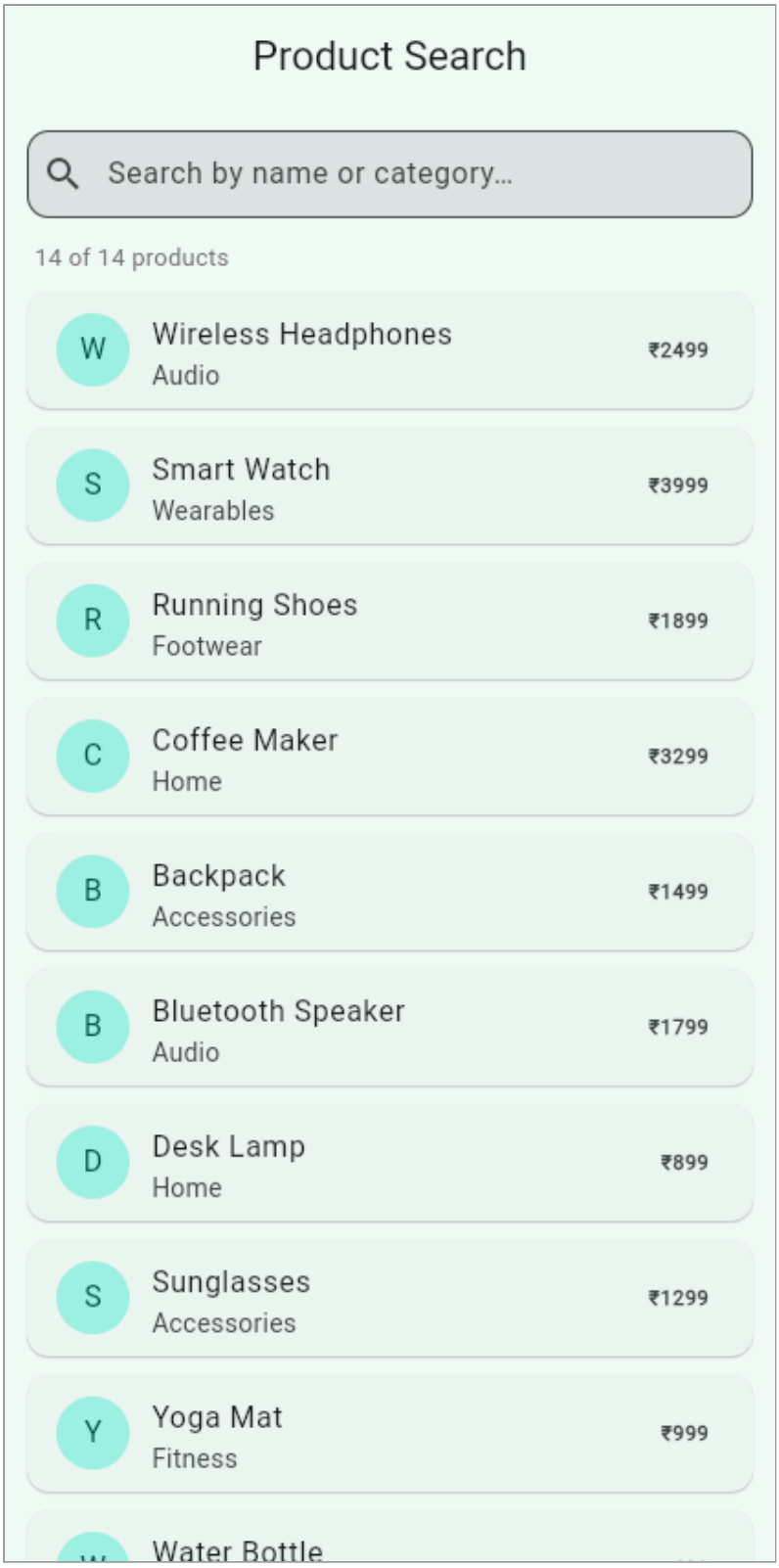


Fig 1. All 14 products shown before any search text is entered.

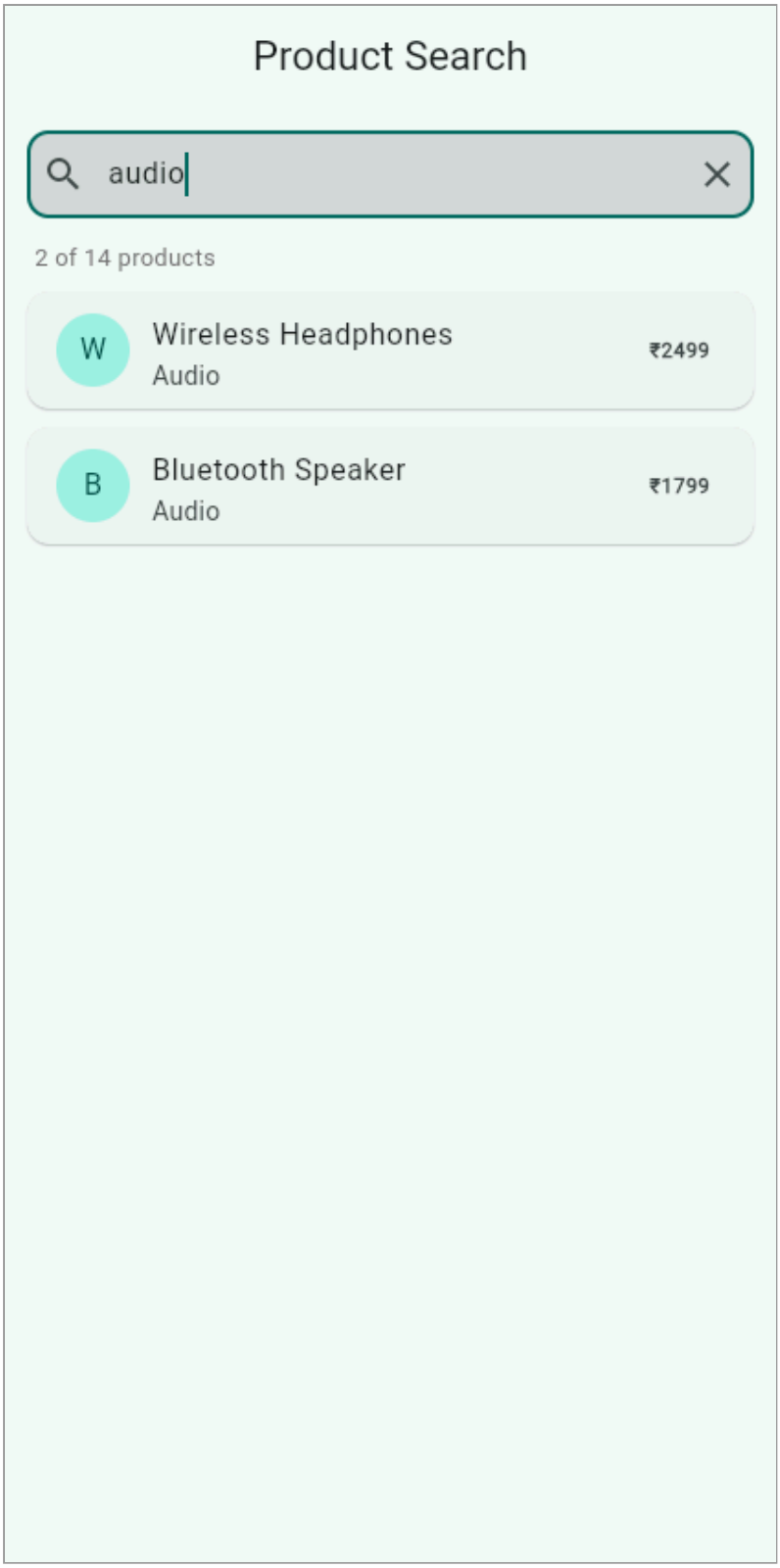


Fig 2. After typing 'audio' into the search field, the list live-filters down to the 2 matching products (Wireless Headphones, Bluetooth Speaker).

Result

The Flutter application was successfully developed to display a predefined list of products and filter them interactively using a TextField input widget. The search correctly narrows the list in real time as

the user types, and an empty-state message is shown when no products match, as verified in the screenshots above.